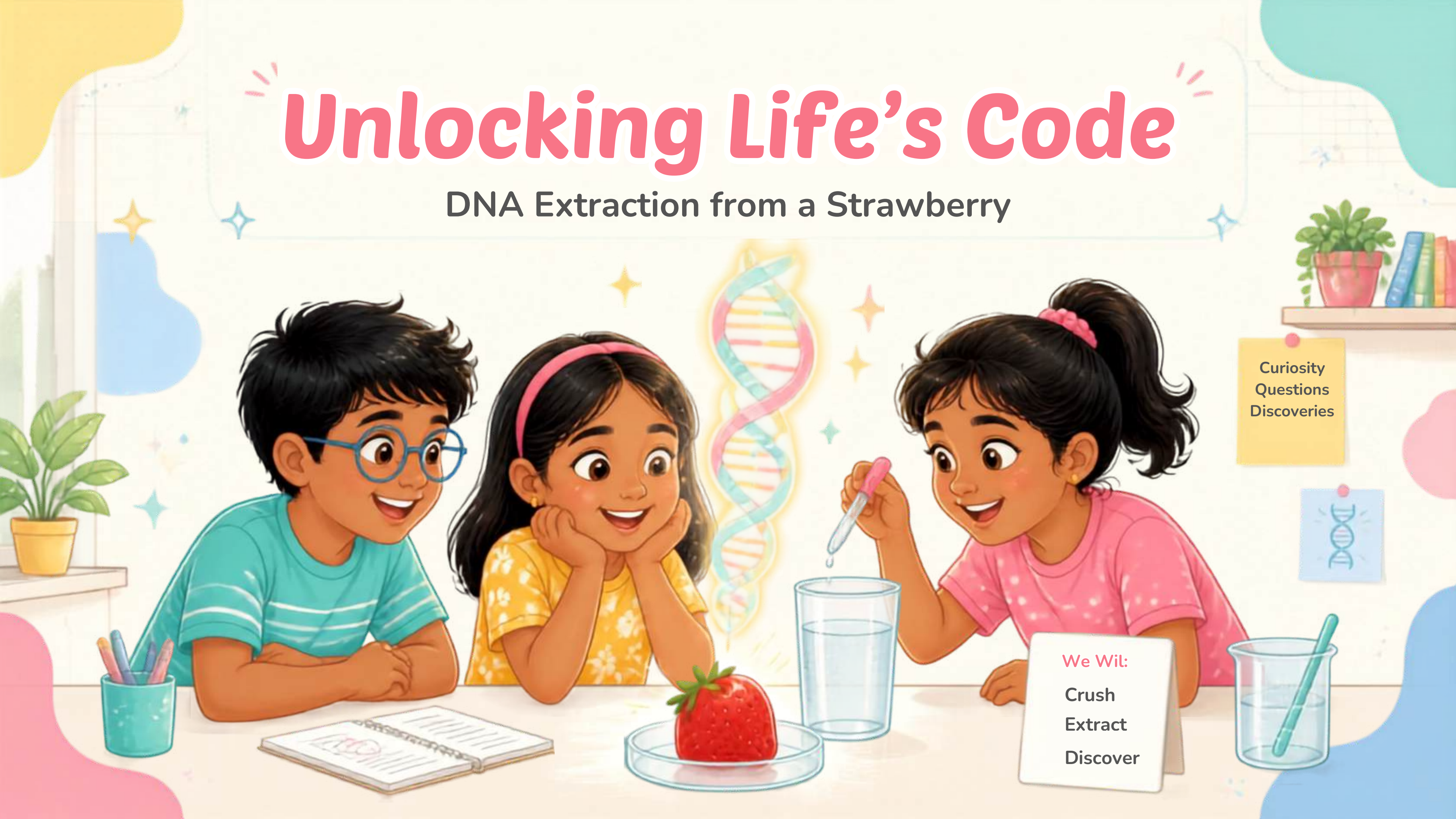


Unlocking Life's Code

DNA Extraction from a Strawberry



Curiosity
Questions
Discoveries

We Will:
Crush
Extract
Discover

What Is DNA?

A tiny recipe book inside every living cell

It helps living things grow and look like themselves.
From a banyan tree to a Bengal tiger, every living thing has DNA.



Gather Your Lab Tools

Strawberry, Ziploc bag, dishwash liquid, salt, water, tea strainer, clear glass, toothpick



Ask an adult to help with cold surgical spirit or rubbing alcohol.



Use kitchen-safe items and stir gently.

1



Strawberry

2



Ziploc bag

3



Dishwash liquid

4



Salt

5



Tea strainer

Clear glass

8



Toothpick



Water

Mash, Mix, Strain

Break the cells, then pour the pink liquid through the strainer



1 Mash the strawberry until it turns pulpy.



2 Add the soap + salt buffer and mix gently.



3 Strain the liquid into a clear glass.



DNA You Can See!

Slowly pour ice-cold alcohol
and watch white threads appear

- ★ Twirl the strands gently with a toothpick.
- ★ Try banana or papaya next and compare the yield.

What's Happening?



DNA is not visible in fruit.
Cold alcohol helps it clump
together so we can see it!

Fruits



Extract



Discover



You'll Need

- Strawberries
- Dish soap
- Salt
- Ice-cold alcohol
- Filter & glass
- Toothpick



Try Banana or Papaya Next

See which fruit gives you more visible DNA



Banana and papaya are easy to find.



Compare the white strands you get.



Share your results with the class!



Compare & Try	Fruit	Easy to Find	White Strands (Amount)	Strands Visible	My Result
 Banana		☆☆☆	☆☆☆	☆☆☆	
 Papaya		☆☆☆	☆☆		

Note

Use the same steps as before. Which fruit gave you more visible DNA?

What Changed?

Different fruits can show different amounts of DNA



Some fruits are softer and easier to mash.



Cold alcohol helps DNA clump together.



Try comparing the layers you see.



BEFORE (MASHED FRUIT)



Banana



Papaya



AFTER (COLD ALCOHOL ADDED)



Banana



Papaya



Look closely! The more DNA, the thicker and cloudier the white layer.

Safety First

Always ask an adult to help with the alcohol step



Use only clean kitchen tools.



Pour slowly and gently.



Wash your hands after the activity.



Ask an adult before the alcohol step.



Alcohol is cold and not for drinking.



Keep your workspace clean and dry.

Quick Check Quiz

Pick the best answer

1. What does soap do?

A Dissolves dirt

B Changes colour

C Makes more DNA

2. Why do we use cold alcohol?

A It melts the DNA

B It helps DNA come out

C It makes bubbles

3. What fruit can you try next?

A Strawberries

B Carrots

C Rice



Think carefully and choose the best answer!

You've got this, *scientist!*

Great Job, Junior Bio-Detectives!

You unlocked the secret code inside a fruit!

